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D. Through the Jungle

time limit per test: 1 second

memory limit per test: 256 megabytes

You are given a **directed unweighted** graph with N nodes and M edges. The nodes are numbered from 1 to N . The graph contains no self-loops or multiple edges.

You have to find a shortest path from node S to node D that passes through node K . If multiple such paths exist, print any one of them. If no such path exists, print -1 .

Input

The first line contains five integers N, M, S, D, K ($1 \leq N \leq 2 \times 10^5, 1 \leq M \leq 3 \times 10^5, 1 \leq S, D, K \leq N$) — the number of vertices, total number of edges, source, destination and the mandatory node that must be included in the path.

The next M lines will contain two integers u_i, v_i ($1 \leq u_i, v_i \leq N$) — denoting there is an edge from city u_i to city v_i .

Output

- If a valid path exists from S to D through K , print the length of the path (number of edges) on the first line.
- On the second line, print the nodes in the path in order from S to D .
- If no such path exists, print -1 .

Examples

input	Copy
<pre> 5 10 5 3 5 2 5 5 1 4 5 3 5 1 2 2 4 3 2 1 4 1 3 3 4 </pre>	
output	Copy
<pre> 2 5 1 3 </pre>	
input	Copy
<pre> 5 4 2 5 3 5 1 3 1 2 3 2 4 </pre>	
output	Copy
<pre> -1 </pre>	
input	Copy

CSE221 Assignment 05 Summer 2026

Contest is running

4 weeks

Contestant



→ Submit?

Language: Java 21 64bit Choose file: Choose File No file chosen

Success!



Submit

```
8 7 3 2 4
7 2
6 7
7 3
2 4
1 2
8 4
5 1
```

output[Copy](#)

```
-1
```

input[Copy](#)

```
6 6 2 2 2
5 1
1 2
1 4
5 2
4 5
3 2
```

output[Copy](#)

```
0
2
```

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